

Michelle C. Bou Rojas

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OBJECTIVE

Computer Science & Engineering student with hands-on experience in AI, data pipelines, and end-to-end solution development.

Passionate about turning complex problems into actionable insights through technology.

EDUCATION

Bachelor's in Computer Science & Engineering with a minor in Project Management

Expected Graduation Date: May 2027

University of Puerto Rico Mayagüez Campus, College of Engineering

TECHNICAL SKILLS

Programming & Web Development: Python, Java, C, C++, HTML, CSS, WordPress, Hostinger, REST APIs, FastAPI, Flask

Database, Data Management & Data Visualization: SQL, CSV Processing, Microsoft Power Platforms, Microsoft Dataverse, Azure Data Services, Databricks, ETL/Data Transformation, Relational Database Querying, D3.js, Google Analytics

AI & Machine Learning: AI Agents & LLMs, Retrieval-Augmented Generation (RAG), Multi-Agent Orchestration, Azure API Management (APIM), Prompt Engineering, Generative AI

PROFESSIONAL EXPERIENCE

Tech@Lilly Generative AI Engineer Intern | Global DIA, Analytics & AI Manufacturing Department

May 2026 – August 2026

Eli Lilly USA – Indianapolis, Indiana

- Designed and deployed a multi-agent LLM orchestrator (Python) coordinating 3 child RAG agents to power an alarm repair agent parsing repair documentation into an ontology graph, retrieving relevant subgraphs for contextual troubleshooting, and visualizing results through an interactive D3.js graph. Integrated with Lilly's internal LLM gateway and Azure API Management, implementing 7 compatibility layers and cutting average engineer time-to-action from ~20 to ~2 minutes.
- Collaborated on an engineering knowledge assistant (Python, FastAPI) that returns answers from technical knowledge bases with traceable source references, achieving a 100% pass rate across defined test cases.
- Used Databricks for data processing and exploration supporting AI-driven applications and built Power BI dashboards to turn complex data into fast access and actionable insights for stakeholders.

Tech@Lilly Analyst Coop Student | MQ and Automation Department

August 2025 – December 2025

Lilly Del Caribe – Carolina, Puerto Rico

- Supported automation initiatives within the MQ and Automation Department by assisting in the design, testing, and optimization of workflows to improve operational efficiency.
- Partnered with cross-functional teams to identify inefficiencies, troubleshoot system issues, and implement streamlined solutions that improved process reliability.
- Gained hands-on experience with automation tools and enterprise systems, contributing to continuous improvement efforts that enhance reliability and compliance.

Strategy & Consulting Intern

May 2025-August 2025

V2A Consulting – San Juan, Puerto Rico

- Conducted strategic analyses to evaluate nonprofit organizational capacity, funding streams, and program expansion opportunities.
- Collaborated with consultants and peers to design actionable recommendations addressing operational efficiency and long-term sustainability.
- Delivered client-ready reports, developed a visual strategy gallery to present findings to nonprofit stakeholders, and produced a comprehensive final report consolidating all analyses and recommendations.

Website Developer & Social Media Manager

August 2023- Present

Dept. of Social Sciences – University of Puerto Rico, Mayagüez

- Built the department's website entirely from scratch using HTML code and WordPress with responsive, user-friendly design; managed content across Facebook, Instagram, and TikTok; and analyzed performance using Google Analytics and platform-specific insights to inform strategic decisions.

EXTRA-CURRICULAR ACTIVITIES

Logistics Division Leader – Quasar UPRM VEX V5

August 2026 - Present

Business Lead & Web Master - Loxodon-1 UPRM

August 2024 – May 2025

Department Advocate - Association for Computing Machinery Women's Chapter

August 2024 – February 2025

Data Analyst - Computational Chemistry Investigation Dept. Chemistry

August 2022 - May 2023

Active Member – Society of Woman Engineers UPRM

August 2022 – May 2023

CERTIFICATIONS

- AT&T 2024 Technology Academy
- NCWIT Georgia Tech PR & Virgin Island Affiliate
- IBM Project Manager
- Google Data Analytics Specialization
- Google Agile Project Management
- Generative AI for Project Manager